(Please write your Enrollment Number) Enrollment No. 47

MID-TERM EXAMINATION
(B. Tech.CSE/CSE-AI/AI&ML/Reppear) (Semester : 4th)
(March, 2024) OFF LINE mode

Subject Code: BCS 204 | **Subject:** Design and Analysis of Algorithms
Time : 1 ½Hours | **Maximum Marks :** 30
Note: Attempt all questions

Q1 **Attempt all parts** (2.5*4)

(a) Solve : $T(n) = T(n^{1/3}) + T(n^{2/3}) + \log_3 n$

(b) Assume that n is a power of 3 in case of Strassen's Matrix Multiplication. Assume that there is a way to multiply 2 matrices of size 3 × 3 with 25 scalar multiplications and 50 scalar additions. What is the complexity of a recursive algorithm that is based on this method?

(c) Given 10 activities along with their start and finish time as :

A(i)	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
S(i)	1	2	3	4	7	8	9	9	11	12
F(i)	3	5	4	7	10	9	11	13	12	14

Compute a schedule where the largest number of activities takes place.

(d) Under a Huffman encoding of N symbols with frequencies $f_1, f_2, f_3, \ldots\ldots, f_N$. What is the longest a codeword could possibly be? Justify your answer.

(5,5)

Q2 **(Attempt any Two Parts) UNIT-1**

(a) Prove that, if $T(N) = xT\left(\frac{N}{y}\right) + N^z$ where $x \geq 1, y \geq 1$ and $z > 0$ then the solution to the recurrence is as follow

$$T(N) = \begin{cases} \theta(N^{\log_y x}) & , x > y^z \\ \theta(N^z \log_y N) & , x = y^z \\ \theta(N^z) & , x < y^z \end{cases}$$

(b) Let an array $A[1:N]$ having integer variables only. $A[1:N]$ is said to have a majority element if more than half of its entries are the same. The task is to design an efficient algorithm with asymptotic complexity of $O(N \log N)$ using divide and conquer strategy to tell whether the array has a majority element, and, if so, to find that element.

(c) Consider a modified binary search algorithm so that it splits the input not into two sets of almost equal sizes, but into three sets of sizes approximately one third. Write the algorithm and find the asymptotic complexity of this algorithm.

(Attempt any Two Parts) UNIT-2 (5,5)

(a) State Knapsack problem and solve the following instance of the knapsack problem for the capacity = 30.

Object	1	2	3	4	5	6	7	8
Profit	60	1	5	40	20	50	10	30
Weight	30	1	1	10	8	100	8	10

(b) Find an optimal parenthesis of a matrix chain product whose sequence of dimensions is: (A=2x3, B=3x4, C=4x5, D=5x6, E=6x7)

(c) Write an algorithm to find longest common subsequence (LCS) and then determine LCS of two given sequences (ABABABA) and (ABAABAAB).